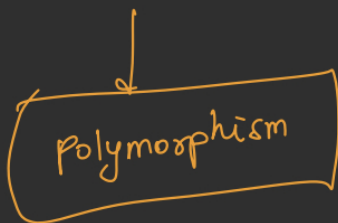
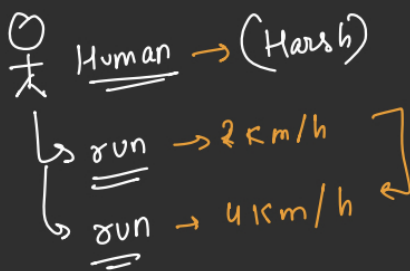


Abstraction
Encapsulation
Inheritance



Polymorphism



Class Human {

```
void run() {  
    → 2 km/h;  
}
```

```
void run(boolean  
isDgBehind) {  
    → 4 km/h;  
}
```

Method Overloading

Compile time
Polymorphism

```
main() {  
    Human *h = new Human();  
    h → run();  
    h → run(true);  
}
```

Polymorphism

Many

Forms

→ One Object can have Many Forms.

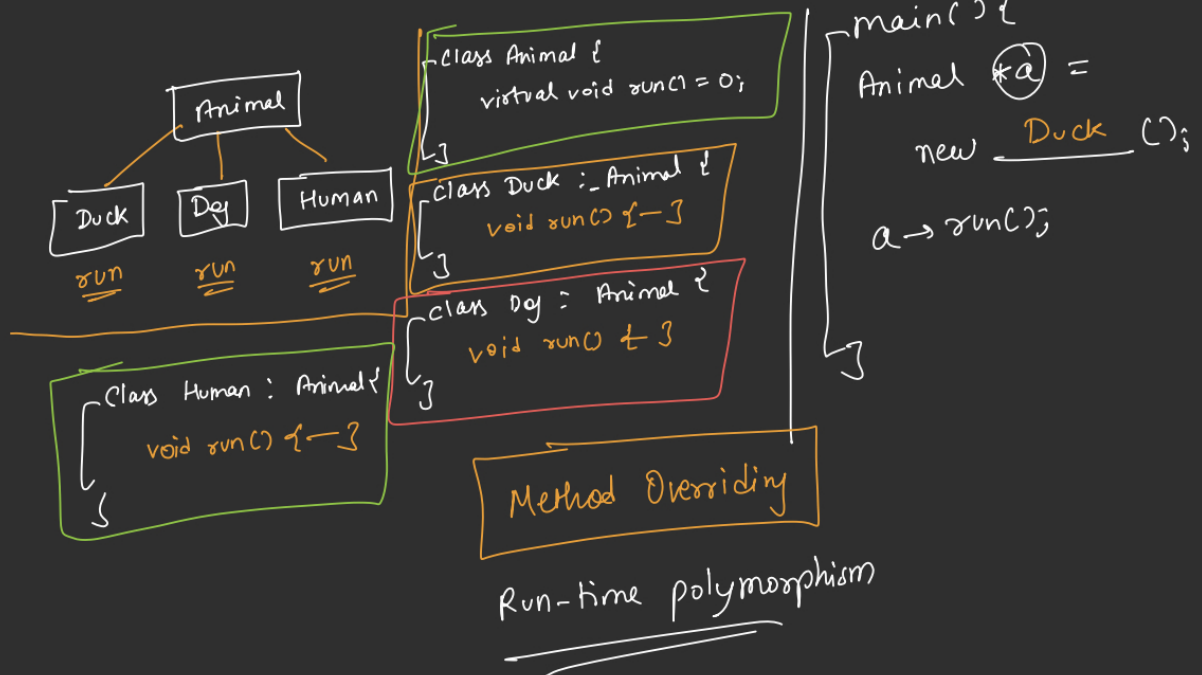
Polymorphism

Compile time polymorphism

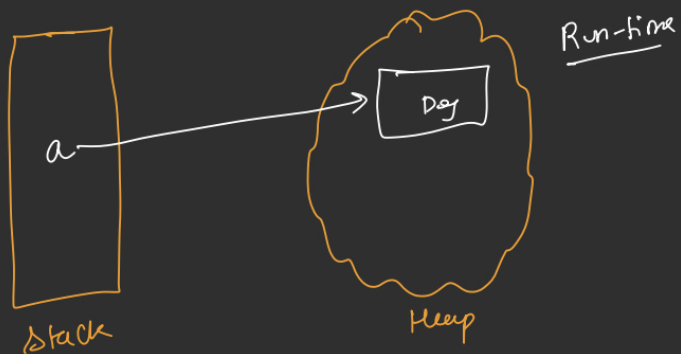
Run-time polymorphism

Method Overloading

Method Overriding



```
Animal *a = new Dog();
           new Duck();
           new Human();
```





OOPS

Static Keyword

```

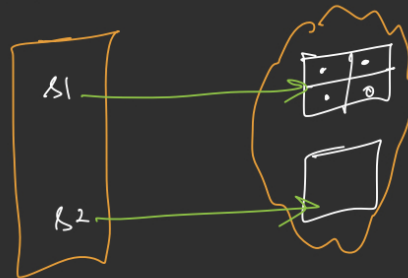
Class Student {
    string name;
    int age;
    int rNo;
    string College;
}
  
```

"DTU"

main() {

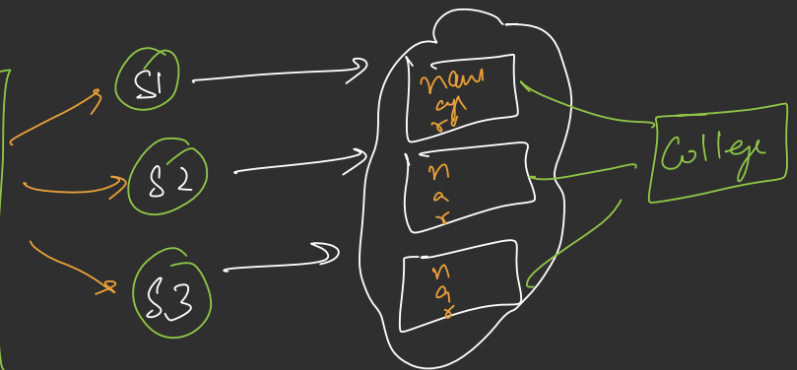
```

Student *s1 = new Student();
Student *s2 = new Student();
  
```



```

Class Student {
    string name;
    int age;
    int rNo;
    Static string College;
}
  
```



s1 → College = 'IITG'

```
class Car {  
    static int noOfWheels;  
}
```

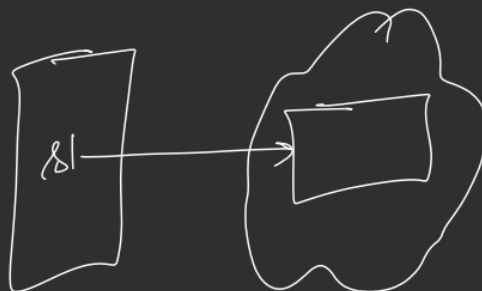
Const →

```
class Student {  
    const static string college;  
}
```

Destructors

↓
Constructor (Object Create)

```
Student * s1 = new Student();
```



delete s1;

class Student {

string name;

int age;

Student() {
// empty.
}

~Student() {
// empty.
}

};

Destructor cannot be overloaded